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SAFETY PLAN GENERATOR

Project: Quick Analysis

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Of course. As STRUCTMIND AI, I will generate a comprehensive safety plan based on the parameters you provided.

Steel Erection Safety Plan: 3-Storey Structure

Project: Generic 3-Storey Steel Erection Project Governing Standard: OSHA 1926 Subpart R – Steel Erection Date: [Date] Prepared By: STRUCTMIND AI

Disclaimer: This is a template safety plan based on the information provided. This document must be reviewed, modified, and approved by a project-specific Competent and Qualified Person prior to the commencement of any work. The site-specific erection plan, crane load charts, and engineered erection drawings shall always take precedence.

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1. Lifting Plan Notes

This section details the plan for the heaviest and most critical lifts. All other lifts shall follow these principles.

1.1. Crane Details:

• Crane Type: Minimum 60-Ton Class Hydraulic Crane (Final selection based on full load chart review).

• Inspections: A valid annual crane inspection certificate must be on site. The operator must conduct and document a daily pre-shift inspection.

• Ground Conditions: Crane setup location must be on firm, level, and compacted ground, verified by a Competent Person. Use of appropriately sized outrigger pads/mats is mandatory.

1.2. Critical Lift Details (Heaviest Pick):

• Load Description: Spandrel Girder

• Estimated Load Weight: 4.2 tons (8,400 lbs)

• Rigging Weight (estimated): 500 lbs (slings, shackles, spreader bar)

• Crane Block/Ball Weight: 350 lbs

• Total Calculated Load: 9,250 lbs

• Maximum Lift Radius: 70 ft

• Capacity Check: The crane's load chart must show a capacity significantly greater than 9,250 lbs at a 70 ft radius. Lifts shall not exceed 75% of the crane's rated capacity at that configuration. The operator must verify this before the pick.

1.3. Operational Plan:

• Pre-Lift Meeting: A mandatory pre-lift meeting will be held with the crane operator, signal person, riggers, and erection crew to review this plan, personnel roles, and communication signals.

• Swing Path: The entire swing path shall be cleared of personnel, materials, and equipment. A 10 ft clearance must be maintained from all obstructions.

• Power Lines: All work must maintain a minimum distance of 20 feet from overhead power lines up to 350kV. If closer work is required, lines must be de-energized and visibly grounded.

• Weather Conditions: All lifting operations will cease if wind speeds exceed 20 mph or in the presence of lightning, heavy rain, or other adverse conditions that affect safety. The crane manufacturer's wind speed limits are the absolute maximum.

• Personnel:

• Operator: NCCCO certified or equivalent, and qualified on the specific make/model of crane.

• Signal Person: Qualified per OSHA 1926.1428. Will be the only person signaling the crane unless a "stop" signal is required.

• Riggers: Qualified per OSHA 1926.1425.

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2. Rigging Plan

2.1. General Rigging Requirements:

• All rigging equipment (slings, shackles, hardware) shall be inspected by a Qualified Rigger prior to each shift and each lift.

• Damaged or defective rigging will be immediately removed from service and destroyed.

• Working Load Limit (WLL) tags must be legible on all slings.

• Soften or pad all sharp edges on steel members to protect synthetic slings.

2.2. Rigging Configuration for 4.2-Ton Spandrel Girder:

• Method: Two-point pick utilizing a spreader bar to ensure vertical sling legs onto the load, preventing inward crushing forces.

• Components (Minimum WLL):

• Spreader Bar: Rated for a minimum of 5 tons (10,000 lbs).

• Top Slings (Hook to Spreader Bar): Two (2) wire rope or synthetic slings. Sling angle shall not be less than 60 degrees. WLL must be sufficient for the angled load.

• Bottom Slings (Spreader Bar to Girder): Two (2) wire rope or synthetic slings of equal length, attached to engineered lift points or choked around the girder. Each sling must have a vertical WLL of at least 2.5 tons (5,000 lbs).

• Shackles: Four (4) bolt-type anchor shackles with a minimum WLL of 3.25 tons. Ensure pins are properly seated and secured.

• Load Control: A minimum of two (2) non-conductive tag lines will be used to control the load's movement. Tag line handlers will remain on the ground, clear of the load's direct path.

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3. Fall Protection Plan (per OSHA 1926 Subpart R)

3.1. General Requirement:

• All personnel working at a height of 15 feet or more above a lower level must be protected from falls by conventional fall protection systems (guardrails, personal fall arrest systems, etc.), unless otherwise specified below.

3.2. Connectors:

• Connectors working between 15 feet and 30 feet (or two stories, whichever is less) may work without being tied off only while actively connecting and moving between connection points.

• Connectors must be 100% tied off at all times when working above 30 feet (or two stories).

• A dual-leg, 100% tie-off lanyard or Self-Retracting Lifeline (SRL) is required. Acceptable anchor points must be capable of supporting 5,000 lbs.

3.3. Decking Crew & Controlled Decking Zone (CDZ):

• A Controlled Decking Zone may be established for steel decking installation.

• CDZ Requirements:

• The CDZ is limited to a maximum area of 90 feet by 90 feet.

• It will be clearly marked with control lines (e.g., flagging, ropes).

• Access is restricted to trained personnel engaged in leading-edge decking work.

• Fall protection is required for deckers at the leading edge over 30 feet (or two stories).

• Decking bundles must be landed on structural members that can support the load.

• Decking must be secured (tack welded or mechanically fastened) as it is installed.

3.4. All Other Personnel:

• Welders, bolters, detailers, and any other personnel on the erected steel must be protected by a Personal Fall Arrest System (PFAS) at all times when working at heights of 15 feet or more.

• Perimeter safety cables will be installed as soon as the decking is in place to provide a guardrail system. Cables will be set at 42 inches (top rail) and 21 inches (mid-rail).

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4. Erection Stability & Bracing Plan

4.1. Column Anchorage:

• All columns will be erected on a 4-anchor rod system.

• The controlling contractor must provide written notification that the concrete/grout has achieved sufficient strength per the Engineer of Record (EOR).

• Columns must be braced with a minimum of two (2) temporary guy wires/cables or rigid braces before the crane load line is detached.

4.2. Connections:

• Beams & Girders: Each structural member must be secured with a minimum of two (2) bolts per connection, drawn up wrench-tight, before the crane is unhooked.

• Joists: Where joists span between beams, they must be bolted or welded to the structure to create a stable bay before placing any construction loads (e.g., decking bundles) on them.

• Diagonal Bracing: Temporary and permanent bracing specified in the erection drawings must be installed concurrently with the main structural members to ensure frame stability. No temporary bracing shall be removed without approval from the Competent Person and/or EOR.

4.3. Erection Sequence:

• Erection will proceed in a planned sequence as detailed in the engineered erection drawings to maintain structural stability at all times.

• Per Subpart R, there will be no more than four (4) floors of unfinished bolting or welding above the foundation or the highest fully planked floor.

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5. Hazard Register & Control Measures

Hazard	Potential Harm	Control Measures
Crane Overturning	Catastrophic collapse, multiple fatalities	Proper ground prep, use of outrigger pads, strict adherence to load charts, adherence to wind speed limits, certified operator.

Hazard	Potential Harm	Control Measures
Dropped Load	Struck-by fatality/injury, property damage	Qualified rigger, daily rigging inspection, use of tag lines, established "No Go" zones under suspended loads, clear communication.
Falls from Height	Severe injury or fatality	100% adherence to the Fall Protection Plan (Section 3), proper training on PFAS use, inspection of all fall protection equipment.
Structural Collapse	Catastrophic failure, multiple fatalities	Adherence to the Stability & Bracing Plan (Section 4), following erection sequence, ensuring bolts are installed before unhooking, Competent Person oversight.
Electrocution	Burns, fatality	Maintain >20 ft clearance from power lines. De-energize/ground lines if clearance cannot be maintained. Use non-conductive tag lines.
Struck-by/Caught-in	Impact injuries, crushing, amputations	Controlled access zones, no work under suspended loads, situational awareness, clear communication between signal person, operator, and crew.
Material Handling	Strains, sprains, pinch points	Proper manual lifting techniques, use of mechanical aids where possible, wearing appropriate gloves.

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6. Personal Protective Equipment (PPE) & Site Access

6.1. Minimum PPE: The following PPE is mandatory for all personnel entering the steel erection area:

• Hard Hat (ANSI Z89.1, Type 1, Class E)

• Safety Glasses with side shields (ANSI Z87.1)

• High-Visibility Vest or Shirt

• Steel-Toed Work Boots

6.2. Task-Specific PPE:

• Fall Protection: Full-body harness and appropriate lanyard/SRL for all work at height.

• Rigging/Connecting: Snug-fitting leather gloves.

• Welding: Welding hood, leathers/FR clothing, respirator as required.

• Grinding/Cutting: Face shield worn over safety glasses.

6.3. Site Access & Control:

• The area directly under and around the steel erection activities is a Controlled Access Zone.

• Only essential personnel (the erection crew) are permitted inside this zone.

• Signage will be posted at all access points warning of "Overhead Crane and Erection Work."

• All personnel must check in with the site supervisor or Competent Person before entering the work area.